

ZACHARY HEATHCOCK

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EDUCATION

University of Memphis

August 2021 - May 2025

Bachelor's, Computer Engineering

GPA: 3.79

- Computer Organization and Computer Architecture, Discrete Structures, Digital Circuit Design, Intro to Microprocessors

SKILLS

Hardware Design: Magic, NgSpice, Verilator, Oscilloscope, Assembly | **Simulation:** Kicad, PSpice, MATLAB, Simulink, LTSpice, Cadence Allegro | **Coding:** Python, Git, VBA | **Embedded:** Linux, PuTTY, WinSCP, SSH, Bash Scripting, Arduino

CERTIFICATIONS

Electrical Engineer in Training, 2025 NCEES TN FE Exam

PROFESSIONAL EXPERIENCE

Peak+

May 2025 - May 2026

Embedded Systems Engineer

Memphis, TN, USA

- Led onboarding and commissioning for 180 controllers across 50 sites, serving as the primary engineer for all new system deployments.
- Rolled out improved Modbus/BACnet protocol support to 8 high priority sites, enabling direct Ethernet data readout by third-party building automation systems and eliminating the need for intermediary hardware.
- Partnered with the database engineer to streamline commissioning data pipelines, cutting controller onboarding time by 18% and reducing manual data handoffs.
- Co-authored internal documentation standardizing the controller onboarding and commissioning process, enabling consistent and repeatable deployments across the organization.

Peak+

June 2024 - May 2025

Electrical Engineering Intern

Memphis, TN, USA

- Utilized embedded Linux (Debian) to increase IoT connectivity between system controllers and sensors.
- Developed and expanded a suite of VBA automations to fully replace manual production documentation processes, reducing turnaround time by 44% and eliminating recurring data entry errors.
- Collaborated with sales representatives to design technically sound system configurations within site-specific constraints, ensuring both feasibility and cost-effectiveness during the quoting phase.
- Indexed site performance across 50 sites to identify issues as well as showcase performance based on requests from customers.

PROJECTS & OUTSIDE EXPERIENCE

Orcad PCB Variable gain signal conditioner PCB design

November 2023 - November 2023

Memphis, TN, USA

- Employed Orcad PCB to design the gerber files of a variable gain signal conditioner.
- Analyzed the performance of the design mockup and made changes to the PCB design accordingly.
- Ensured efficient routing channels, placements, and connections to minimize design sizing and behavior.

Intelligent Bi-Directional Energy Controller for Renewable Energy (University of Memphis)

August 2024 - May 2025

Memphis, TN, USA

- Collaborated with a team of fellow students to research, design, and procure a solution with the intention of optimizing energy flow between renewable sources and battery storage systems.
- Led a team of 4 through weekly sprint planning to keep on track for a successful May 2025 demo.
- Applied particle swarm optimization to reduce switching transient noise by 7%, resulting in cleaner output power.